higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

MATHEMATICS LITERACY
(First Paper)
NQF LEVEL 2

5 NOVEMBER 2019

SYMBOL	EXPLANATION
M	Method
MA	Method with accuracy
CA	Consistent accuracy
A	Accuracy
C	Conversion
S	Simplification
RT/RG/RD/RM	Reading from a table/graph/drawing/document/map
F	Choosing correct formula
SF	Substitution in formula
MF	Manipulation of formula
R/J	Reasoning/Justification
P	Penalty, e.g. for no units, incorrect rounding off, etc.
R	Rounding off
E	Explanation

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This marking guideline consists of 7 pages.

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QUESTION 1 (Do not deduct marks if the 'R' sign is omitted.)

QUESTION		SOLUTION	EXPLANATION	
1.1	1.1.1	$\frac{1}{2} + 10 \times 40$ $= \frac{1}{2} + 400$ $= 400\frac{1}{2}$ or 400,5✓	1A	
	1.1.2	$(20 - 6) - (50 - 36)$ $= 14 - 14$ $= 0$✓	1A	
	1.1.3	$8^2 + \sqrt{36}$ $= 64 + 6$ $= 70$✓	1A	
			(3 × 1)	(3)
1.2		19ºC ; 0ºC ; – 4ºC ; – 20ºC ✓ ✓ (One mark if any 2 correct consecutive values)	2A	(2)
1.3		$\frac{20}{25} \times 100$ ✓ ✓ = 80% (Do not deduct marks if % sign is omitted)	1M 1A (2 answer only)	(2)
1.4	1.4.1	08:15✓	1A	(1)
	1.4.2	14:28✓ - 08:15✓ 06:13 The trip took 6 hours and 13 minutes. ✓ **OR** 08:15 – 14:15 = 6 hours✓ 14:15 – 14:28 = 13 minutes✓ The trip took 6 hours and 13 minutes✓ or any other suitable method (3 marks answer only)	CA1.4.1 2M 1A	(3)
1.5		2341 gram ÷ 1000✓ = 2,341 ✓kilogram ≈ 2,3 ✓kilogram (3 marks answer only)	1M 1A 1C	(3)

1.6	1.6.1	2✓: 3✓	2A	(2)
	1.6.2	$\frac{8}{20} \times R500$ or $\frac{2}{5} \times R500$ = R200✓ = R200 (3 marks answer only)	2M 1A	(3)
1.7		$\frac{20}{100} \times R5\,250$ ✓ or $\frac{80}{100} \times R5\,250$ = R1 050 ✓ = R4 200✓ ∴ R5 250 – R1 050 = R4 200✓ (3 marks answer only)	2M 1A	(3)
1.8	1.8.1	R39,75 estimated to R40✓ Therefore R40 × 3✓ = R120✓ (No marks for R119,25 as the answer, it must be an estimate.)	1R 1M 1A	(3)
	1.8.2	R39,75 × 3✓ = R119,25✓ (2 marks answer only)	1M 1A	(2)
1.9		R18,85✓ × 600✓ = R11 310✓ (3 marks answer only)	2M 1A	(3) **[30]**

QUESTION 2

2.1	2.1.1	Cylinder✓	1A	(1)
	2.1.2	$\frac{2,4}{2}$ ✓= 1,2 ✓m (2 marks answer only)	1M 1A	(2)
	2.1.3	$C = 2\pi r$ = 2(3,14)(1,2)✓ = 7,536 m✓ = 7,54 m✓ **OR** $C = \pi d$ = 3,14(2,4)✓ = 7,536 m✓ = 7,54 m✓ (3 marks answer only)	CA 2.1.2 1SF 1A 1R with unit 1SF 1A 1R with unit	(3)

	2.1.4	300 cm = 3 m✓ $V = \pi r^2 h$ ✓ ✓ = (3,14)(2)²(3) = 37,68 ✓ m³	1C 2SF 1A	
		(3 marks answer only)		(3)
2.2	2.2.1	Chips✓	1A	(1)
	2.2.2	E5✓✓	2A	(2)
	2.2.3	South-east✓✓	2A	(2)
	2.2.4	500 000✓ × 4,5✓ 2 250 000 ✓cm 2 250 000 ÷ 100 ÷ 1000✓ = 22,5 ✓km	2M 1A 2C	
		(5 marks answer only)		(5)
2.3	2.3.1	Row A✓ (Note: No mark for A4)	1A	(1)
	2.3.2	2✓	1A	(1)
	2.3.3	Yes✓ There is a picture of a wheelchair indicated on the floor plan.✓	1A 1R/J	(2)
	2.3.4	9 rows✓	1A	(1)
	2.3.5	143✓✓	2A	(2)
	2.3.6	A10✓	1A	(1)
	2.3.7	A = length × breadth = 4✓ × 15✓ = 60 m²✓	 2SF 1A with	
		(3 marks answer only)	Units	(3)
				[30]

QUESTION 3 (Do not deduct marks if the 'R' sign is omitted.)

3.1 3.1.1

INCOME	R	EXPENSES	R
Part-time job	4 500✓	Cell phone	300✓
Pocket money	1 500✓	Loan repayment	1 000✓
		Rent	2 500✓
		Groceries	2 000✓
TOTAL INCOME	6 000✓	TOTAL EXPENSES	5 800✓

8A (8)

	3.1.2	Fixed✓ It is the same every month/It does not change from month to month.✓	1A 1R/J	(2)
	3.1.3	R6 000 – R5 800✓ = R200✓ (2 marks answer of R200 only) surplus✓ (1 mark for surplus only)	CA 3.1.1 1M 2A	(3)
	3.1.4	Save the money. Pay off the loan. ✓ (Any other logical/reasonable answer)	1A	(1)
3.2	3.2.1	2c ✓	1RT	(1)
	3.2.2	Yes✓ 2c × 60c = 120c or R1,20✓	1A 1A	(2)
	3.2.3	(3 minutes × R1,20)✓ + (2c × 20 seconds)✓ = R3,60 + 40c = R4,00✓ OR 3 minutes and 20 seconds = 200 seconds 200 seconds✓ × 2c✓ = 400c or R4,00✓ (3 marks answer only)	2M 1A	(3)
	3.2.4	No✓	1A	(1)
	3.2.5	$\frac{15}{100} \times$ R2 600 ✓ = R390✓ R2 600 + R390✓ = R2 990✓ **OR** $\frac{115}{100}$ ∴ R2 600 ✓ = R2990✓✓ (4 marks answer only)	1M 1A 1M 1A	(4)
3.3	3.3.1	R941,32 Cr	1A	
	3.3.2	365TB86	1A	
	3.3.3	Credit amount/Money coming into the account/Credit received	1A	
	3.3.4	R80,50 (no marks for R80,5)	1A	
	3.3.5	R970,85Cr	1A (5 × 1)	(5)

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QUESTION 4

4.1 4.1.1

Time in hours	0	1	2✓	3✓	4✓	5✓
Number of bottles	0	200✓	400	600✓	800✓	1000✓

8RG (8)

4.1.2 Time in hours✓✓ 2A (2)

4.1.3 Number of bottles✓✓ 2A (2)

4.1.4 Direct relationship✓ 1A
As the time increases,✓ the number of bottles being filled will also increase. ✓ 2R/J (3)

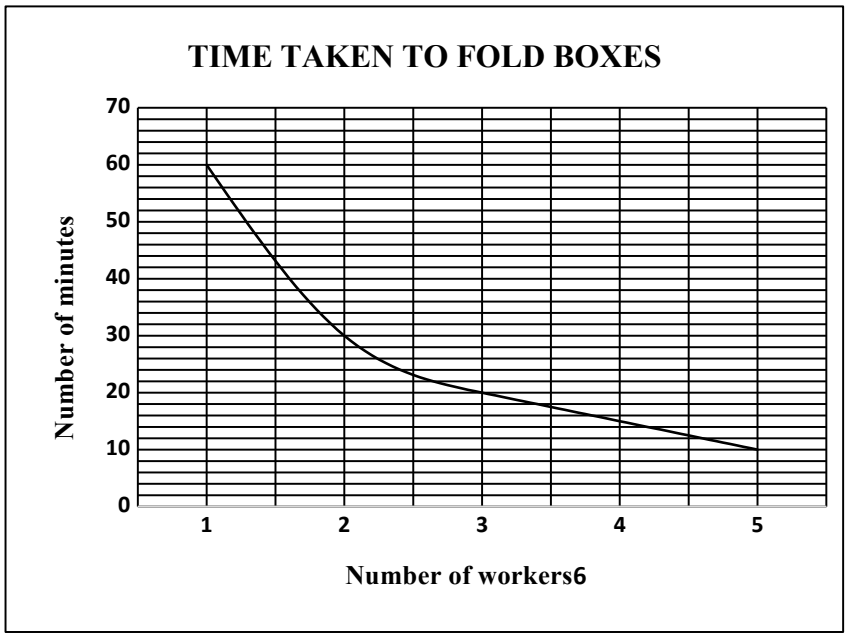
4.2 4.2.1 20 minutes✓ 1RT with minutes (1)

4.2.2 60 ÷ 30✓ 1M
= 2 ✓workers 1A
(2 marks answer only) (2)

4.2.3 60 ÷ 4✓ 1M
= 15✓minutes 1A
(2 marks answer only) (2)

4.2.4 Number of workers✓✓ 2A (2)

4.2.5



CA 4.2.2 and 4.2.3

1 mark (1;60)✓
1 mark (5;12)✓
1 mark correct graph✓ (5)

4.2.6 Inverse/Indirect✓ 1A
As the number of workers increase,✓ the time to complete the task will decrease.✓ 2R/J (3)

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[30]

QUESTION 5

5.1 5.1.1

SCHOOL SUPPLY ITEM	TALLY	FREQUENCY
Binder	𝍸 𝍸 ||||	**14✓**
Notebook	𝍸 𝍸 ||	**12✓**
Highlighter	𝍸 ||||	**9✓**
Folder	𝍸 ||	**7✓**
Crayon	𝍸 ||	**7✓**
Clipboard	𝍸 |	**6✓**
	TOTAL:	**55✓**

7RG

Total: CA with one mistake

Both the tally and the frequency must be correct to get 1 mark. (7)

5.1.2 14✓ 1RT/RG (1)

5.1.3 Clipboard✓ 1RT/RG (1)

5.1.4 Folder✓ and crayon✓ 2RT/RG (2)

5.1.5 14 + 12 + 9 + 7 + 7 + 6✓ 1M
= 55✓ (2 marks answer only) 1A (2)

5.1.6 R25✓ × 12✓ 1RT/RG
R300✓ 1M
(3 marks answer only) 1A (3)

5.2 5.2.1 645✓ ÷ 10✓ 2M
= 64,5✓ 1A
Also accept 64 and 65 as a final answer.
(3 marks answer only) (3)

5.2.2 72✓ – 54✓ 2M
= 18✓ 1A
(3 marks answer only) (3)

5.2.3 62✓ and 68✓ 2A (2)

5.2.4 $\frac{62+68}{2}$ 2M

= 65✓ 1A
(3 marks answer only) (3)

5.2.5 Yes✓ 1A
You could see/observe the contestants eating the hotdogs.✓✓ 2R/J
(Any other suitable answer) (3)
[30]

TOTAL: 150

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